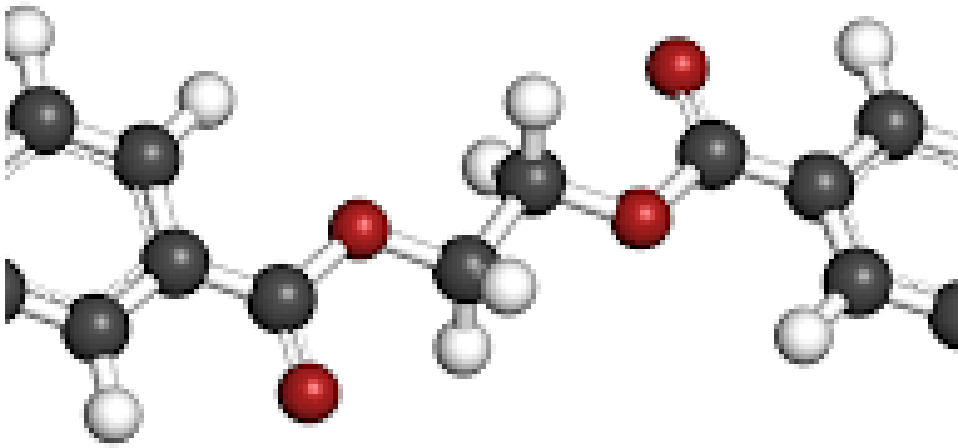
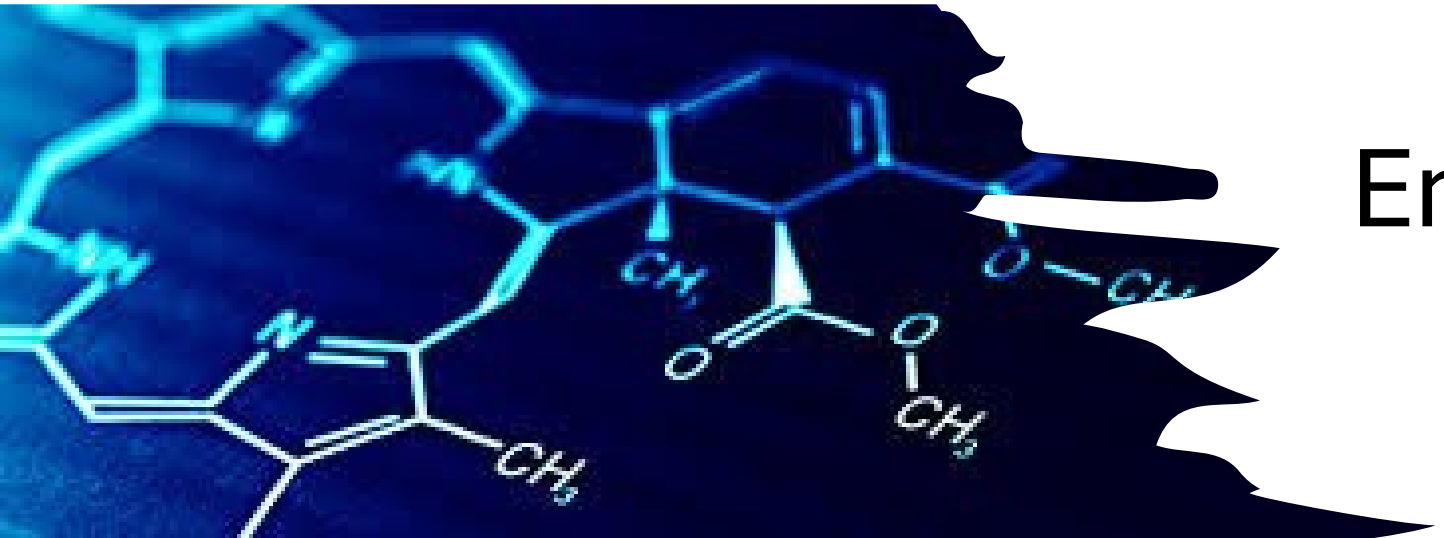




Engineering POLYMERS

Introduction

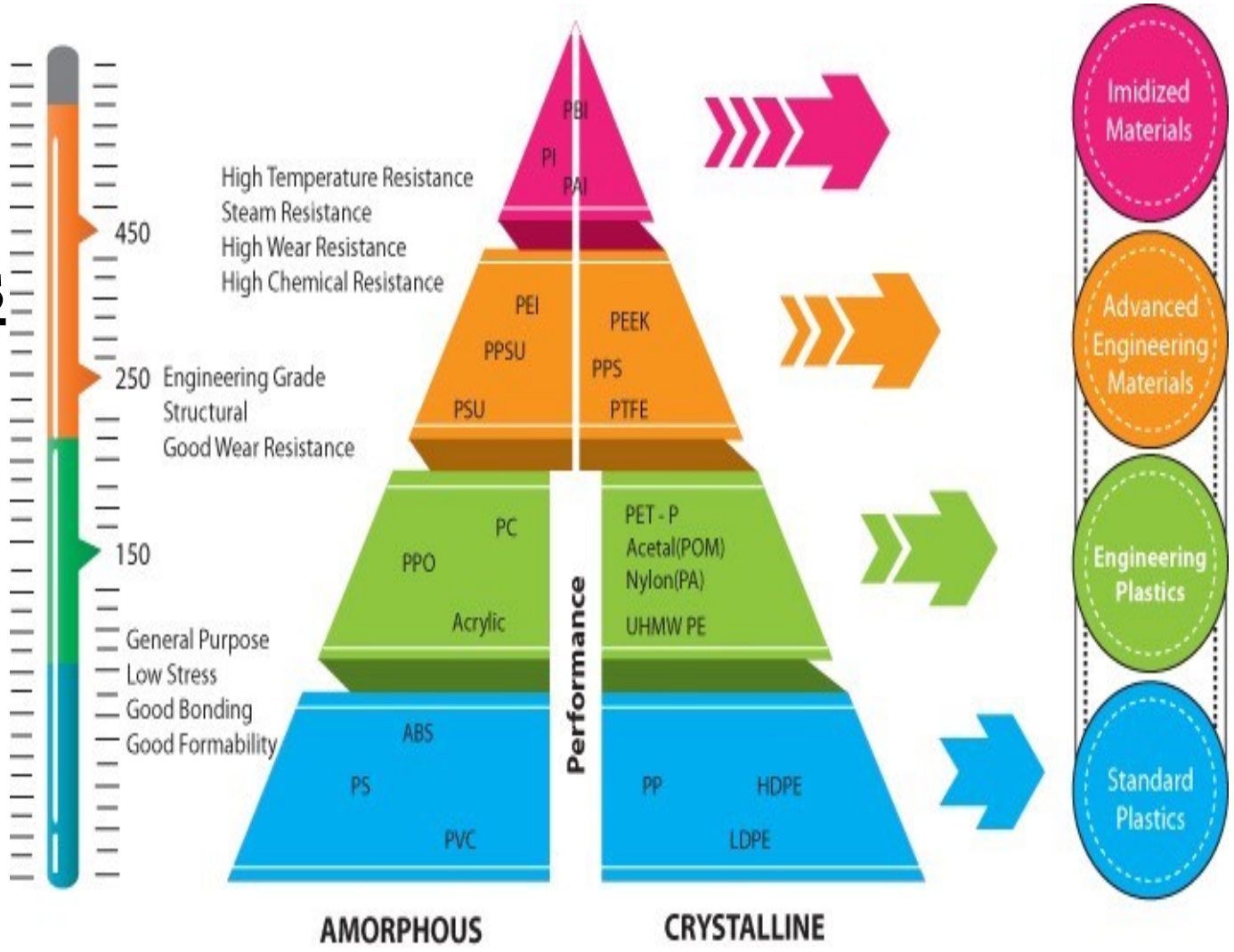
Engineering POLYMERS

Properties

- ✓ Heat resistant
- ✓ Strong
- ✓ Flexible

Used in Industries such as

- ✓ Automotive
- ✓ Aerospace
- ✓ Electronics
- ✓ healthcare



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Definition: Polymer engineering comprises the engineering discipline, focusing on the design, analysis, and modification of the polymer materials.

Includes various aspects like:

- Polymerization Processes
- Polymer Structure and Characterization
- Polymer Properties
- Compounding (mixing polymers with other materials)
- Processing Techniques
- study of major polymers
- structure-property relationships
- Industrial applications

Engineering POLYMERS

Key attributes to Engineering Polymers

Engineering Polymers	Properties	Attributes	Examples
	Mechanical Properties	high strength, stiffness, toughness, and impact resistance	PE, Kelvar, Carbon Fibres, nylon
	Thermal Stability	withstand prolonged exposure to elevated temperatures without significant degradation	polybenzoxazoles (PSPBOs), poly phenyls
	Chemical Resistance	Resistant to chemicals, solvents, and environmental conditions	low density polyethylene (LDPE), Polytetrafluoroethylene (PFTE).
	Electrical Properties	Ability to delocalize π-electrons; Insulation Properties	polyaniline (PANI), Polypyrrole (PPy) silicone, Ethylene Propylene Diene Monomer (EPDM)
	Dimensional Stability	ability to maintain their size even in changing environmental conditions	Polyether ether ketone (PEEK), Polyethylene terephthalate (PET), polyvinylidene difluoride (PVDF)
	Flame Retardancy	Flame resistance	Polyimides, polybenzoxazoles (PBOs), polybenzimidazoles, and polybenzthiazoles (PBTs)

Polyamides

- Common name: Nylon (aliphatic polyamides)
- Preparation: Melt poly condensation process
- Precursors: di-carboxylic acid and diamines

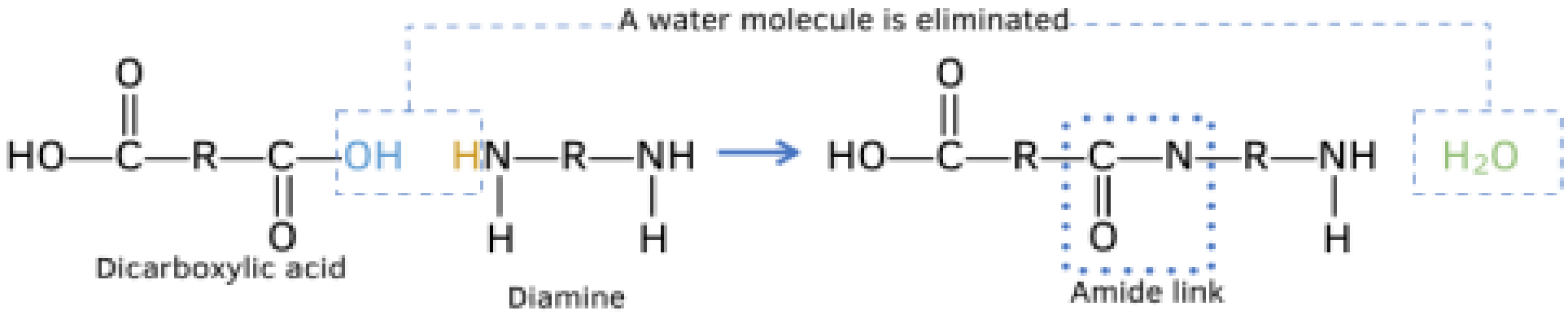


Fig 1. Condensation Reaction between a Dicarboxylic Acid and a Diamine to form a Polyamide.

- synthetic polymers characterized by the presence of repeating amide (–CONH–) linkages along the polymer chain
- amide groups contribute to the **material's strength, toughness, and chemical resistance**

Types of Polyamides

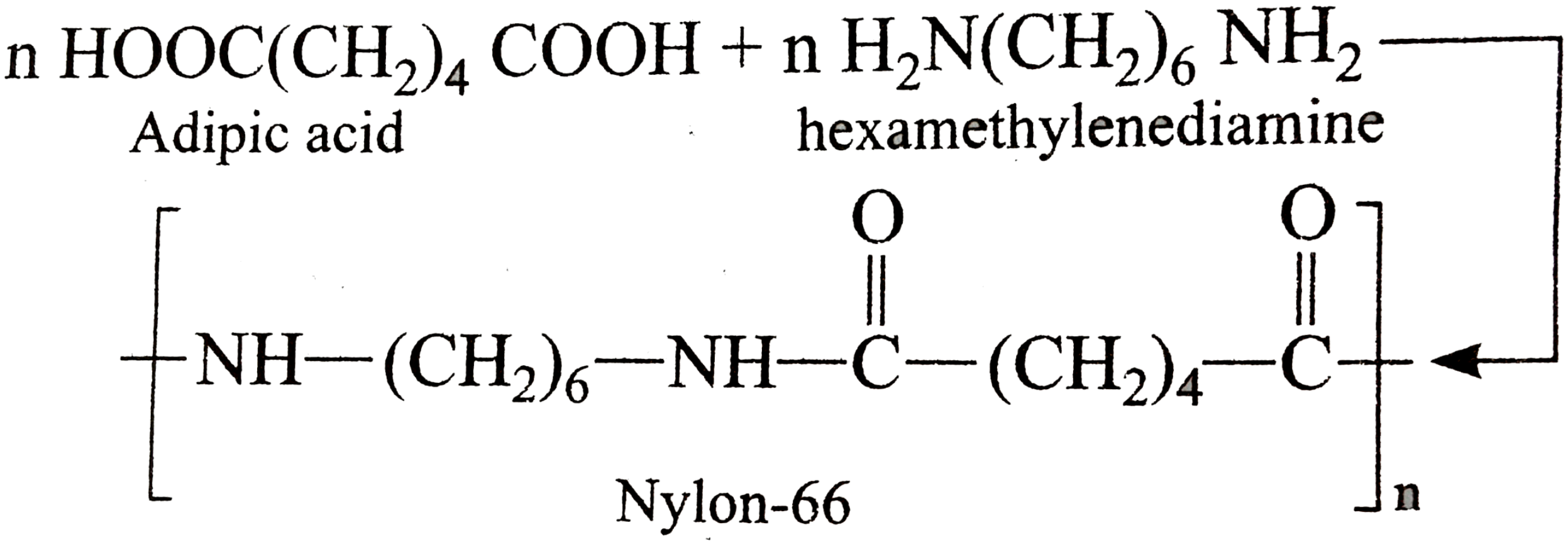
- Types based on number of the carbon atoms present in the monomer molecules.

Examples:

- Nylon 6 (polycaprolactam)
- Nylon 6,6 (polyhexamethylene adipamide)
- Nylon 11 (polyamide 11)
- Nylon 12 (polyamide 12)

Commercially, Nylon 6,6 and nylon 6,10 are extremely important.

Synthesis of Nylon 6,6



Applications of Nylon 6,6

- Nylon 6,6 is used as plastic as well as fiber.

✓ good tensile strength,
✓ abrasion resistance
✓ toughness up to 150 ˚C.
✓ resists many solvents. (However, formic acid, cresols and phenols dissolve this polymer).

- It is used in the production of tyre cord, monofilaments, and ropes.
- Used in the manufacturing of textile fibers.
- good substitute for metals in gears and bearings
- Used to manufacture articles like brushes.

Nylon Fibres

Nylon textile Fibres

Nylon brushes

Plastic Gears

Polyesters

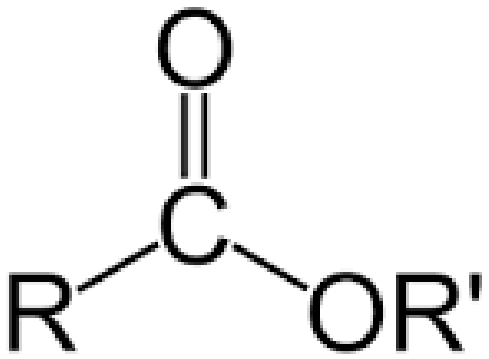
Preparation:

polycondensation reaction between diacids and diols. They have ester functional group in their main chain.

Applications include textiles, packaging etc.

Their properties includes:

- ✓ strength,
- ✓ durability,
- ✓ resistance to shrinking
- ✓ stretching.



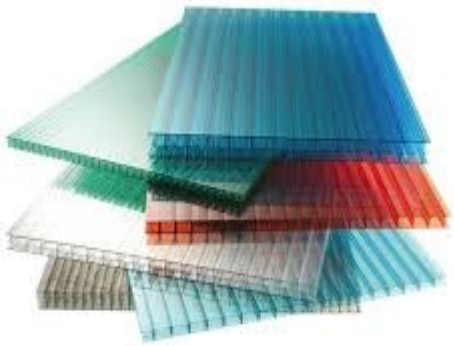
The strength and stiffness of a polyester fiber can be controlled by varying: length of the chains and the degree of crystallinity

However, traditional polyester is derived from petrochemical sources and is not biodegradable, contributing to plastic pollution and environmental degradation.

Types of Polyesters

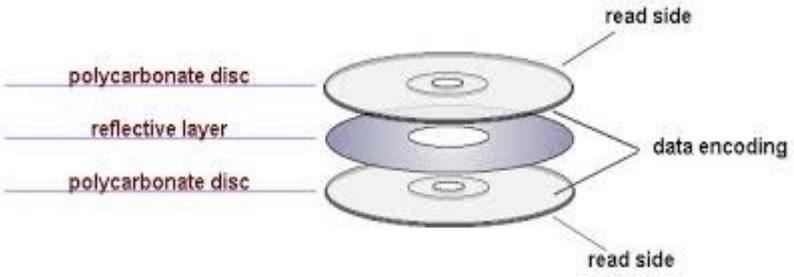
- specific arrangement of the atoms in a polyester molecule can affect its properties

Polycarbonate (PC)



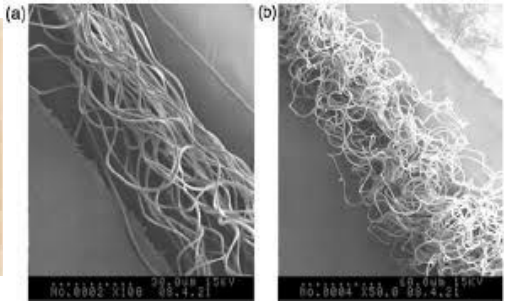
Bullet Proof Glasses

Eye lens



Polytrimethylene Terephthalate (PTT)

Carpets



textiles

In Apparel

Types of Polyesters

Polycyclohexylenedimethylene Terephthalate (PCT)

✓ Electrical connectors, switches, and sockets- **Celanese ™**

For Medical components

Polyester Polyols

polyurethane foams → Flexible foams → For cushions

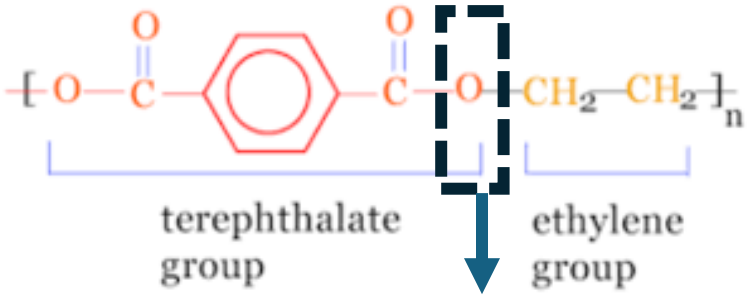
Elastomers → Rigid foams → For insulation, adhesives and coating

All Polyesters has unique properties and has wide range of applications

Polyesters

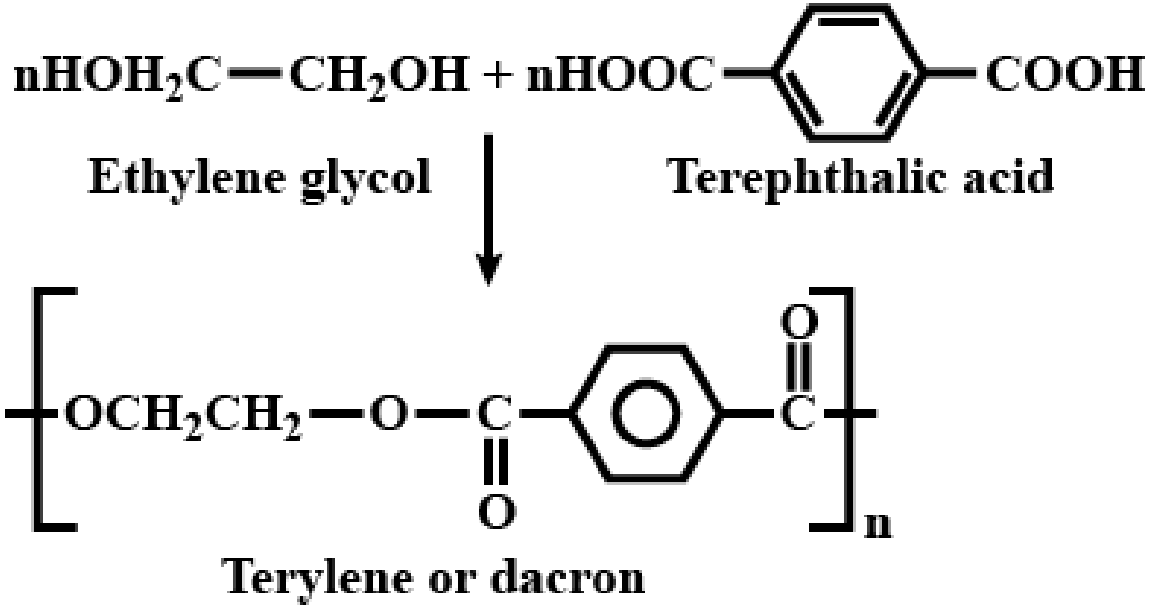
- common type of polyester**: polyethylene terephthalate (PET)**
- Finds application in: clothing, food packaging, and plastic bottles.
- PET is made from the reaction of ethylene glycol and terephthalic acid.

Structure of Polyethylene Terephthalate



Ester linkage

- Synthesis:

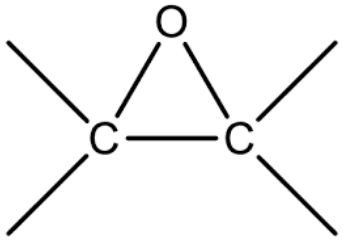


PET resins

PET bottles

Epoxies

- class of reactive prepolymers and polymers which contain epoxide groups

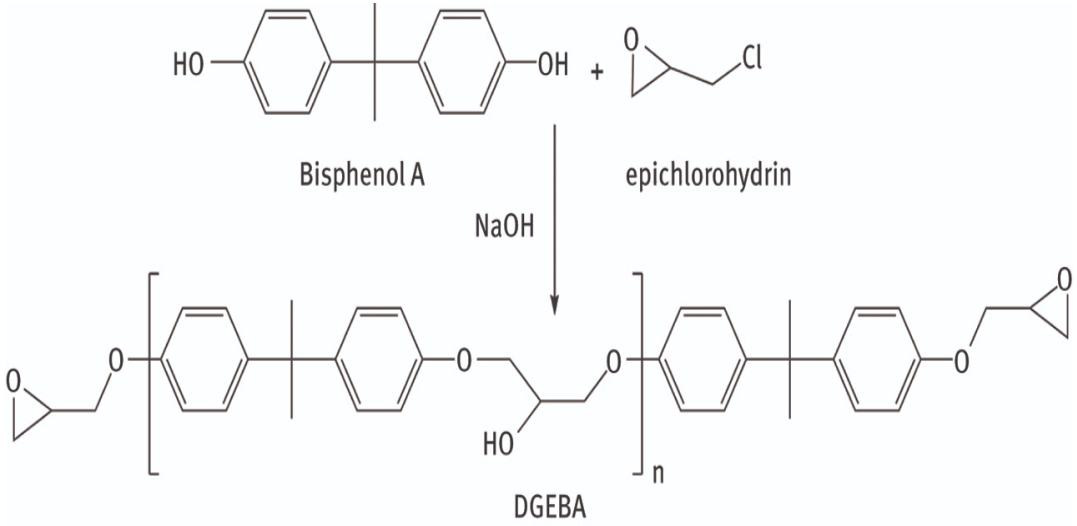


Epoxy group

They possess:

- mechanical properties,
- strong adhesion,
- resistance to chemical and environmental degradation

Synthesis:



Absence of ester groups means that the epoxy resin has particularly good water resistance

Types of Epoxies

1. Bisphenol A Epoxy Resins

Properties

- ✓ High mechanical strength,
- ✓ good thermal chemical resistance
- ✓ excellent adhesion

Uses

- ✓ Coating,
- ✓ Adhesives
- ✓ Electrical and electronic components
- ✓ Composites
- ✓ Laminates

Types of Epoxies

2. Novolac Epoxy Resins

Properties

- ✓ Excellent Mechanical Properties,
- ✓ Superior chemical resistance
- ✓ improved performance at high temperatures

Uses

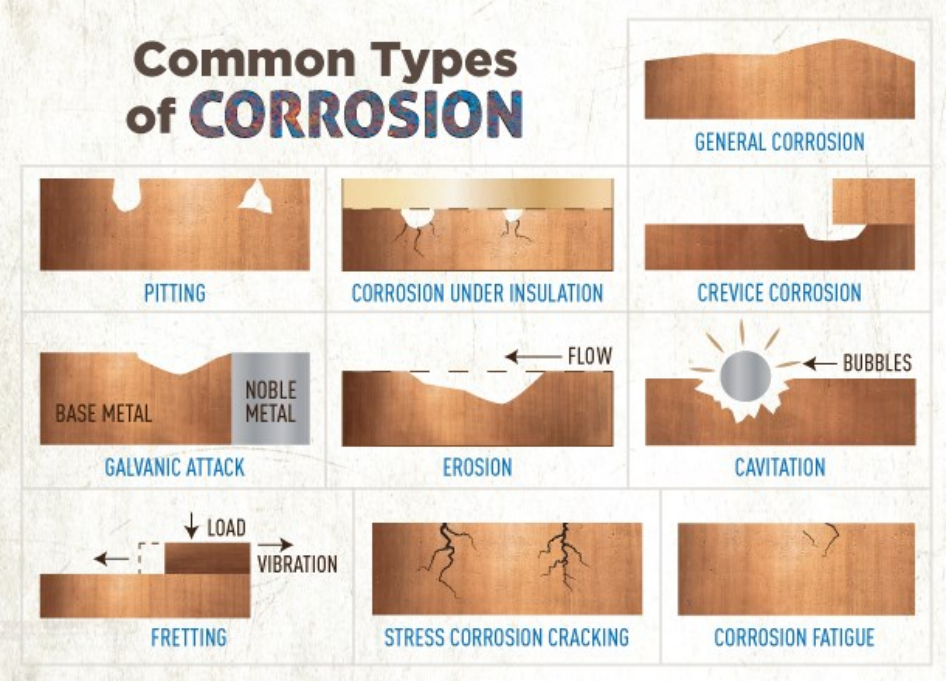
- ✓ Chemical-resistant coatings,
- ✓ high-performance composites,
- ✓ applications requiring high thermal stability

General Properties of Epoxies

- High bonding strength to a variety of substrates including metals, ceramics, glass, and plastics

General Properties of Epoxies

- **Excellent protective properties, resistance to corrosion, chemicals, and abrasion**



Abrasion resistant epoxy

Chemical resistant epoxy

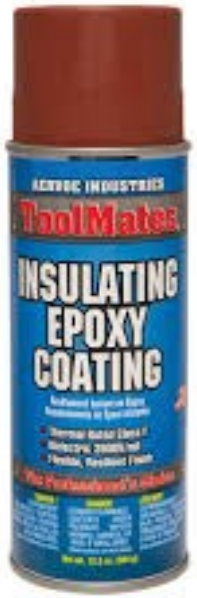
General Properties of Epoxies

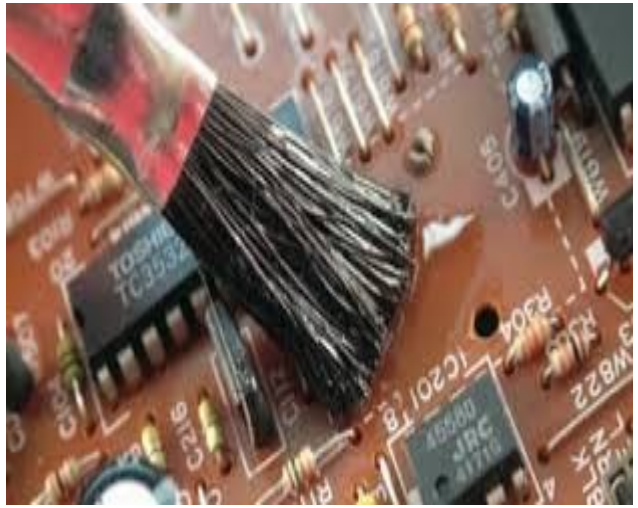
- **High strength-to-weight ratio, durability, and resistance to environmental factors**

Property	Epoxy	Polyester
Viscosity at 25 °C μ (cP)	12000-13000	250-350
Density ρ ($g.cm^{-3}$)	1.16	1.09
Heat Distortion Temperature HDT (°C)	50	54
Modulus of elasticity E (GPa)	5.0	3.3
Flexural strength (MPa)	60	45
Tensile strength (MPa)	73	40
Maximum elongation (%)	4	1

General Properties of Epoxies

- **Excellent insulation properties, thermal stability, and protection against moisture and contaminants**





General Properties of Epoxies

Clear, glossy finish, and ability to embed object

References

- Polymer Science, V R Gowariker, N V Viswanathan, Jayadev Sreedhar, New Age International Publishers, 3rd Edition, 2019
- Green Polymeric Materials: Advances and Sustainable Development, Shakeel Ahmed, Ph.D., Annu Ikram, Saiqa Ikram, Nova Science Publishers, August 2017